



**TRINITY GRAMMAR SCHOOL**  
**MATHEMATICS DEPARTMENT**



**YEAR 12 2009 ASSESSMENT TASK 2**

# **MATHEMATICS**

**(2 UNIT/EXTENSION 1)**

**Time Allowed – *one hour***

**(2 UNIT/EXTENSION 1)**

**Weighting 20% towards final result**

Outcomes referred to: P2, P4, P5, P6, P7, P8, H1, H2, H4, H5, H6, H7, H8, H9

**INSTRUCTIONS:**

1. Attempt ALL questions.
2. Show all necessary working.
3. Begin each question on a new page.
4. Each question is of equal value. Mark values are shown beside each part.
5. Non-programmable silent Board of Studies approved calculators are permitted.
6. If requested, additional writing sheets may be obtained from the examinations supervisor upon request.
7. A double sided A4 page of notes is permitted to be referred to throughout this task.

**Question 1:** A(2,-2), B(-2,-3) and C(0,2) are the vertices of a triangle ABC.

- (a)
- |       |                                                             |   |
|-------|-------------------------------------------------------------|---|
| (i)   | Draw a sketch diagram of the triangle.                      |   |
| (ii)  | Find the length of AC and the gradient of AC.               | 2 |
| (iii) | Find the equation of AC in the general form.                | 2 |
| (iv)  | Calculate the perpendicular distance of B from the side AC. | 2 |
| (v)   | Hence, or otherwise, find the area of $\triangle ABC$ .     | 1 |
- (b) Find the co-ordinates of the point on the curve  $y = 3x^2 - 2x + 1$  where the tangent is parallel to the straight line  $4x - y - 1 = 0$ .
- 3

**Question 2:**

- (a) For the curve  $f(x) = x^4 - 4x^3$ , find the values of  $x$  for which the curve is concave up.
- 2
- (b) Consider the curve  $y = 3x^4 - 8x^3 + 6x^2$ .
- |       |                                                      |   |
|-------|------------------------------------------------------|---|
| (i)   | Find the co-ordinates of its stationary points.      | 3 |
| (ii)  | Use the second derivative to determine their nature. | 3 |
| (iii) | Find any points of inflexion.                        | 2 |
| (iv)  | Sketch the curve for the domain $-1 \leq x \leq 2$   | 3 |

**Question 3:**

(a) Find the primitive functions of the following:

(i)  $3x^5 - 2x^2 - x$

2

(ii)  $3(3x-6)^4$

2

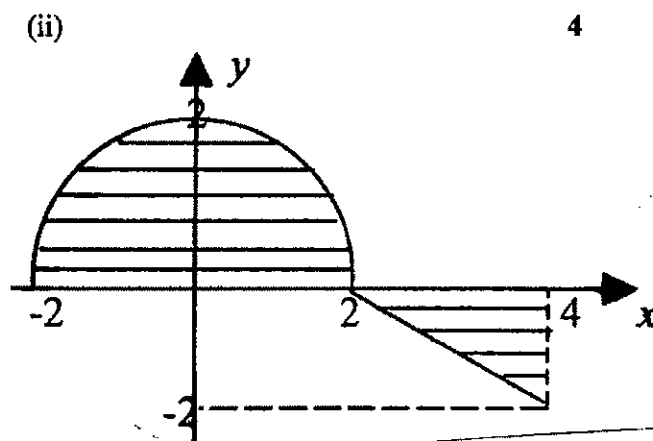
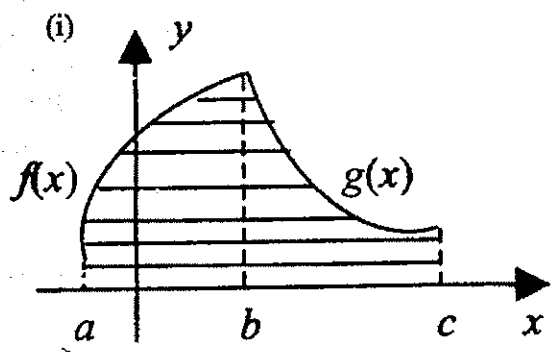
(iii)  $\frac{2}{\sqrt{x}} + 2\sqrt{x} - 3$

3

(b) Express  $y$  in terms of  $x$ , given that  $\frac{d^2y}{dx^2} = 12(x-1)^2$  and that at  $x=1$ ,  $\frac{dy}{dx} = 3$  and  $y = 2$ .

3

(c) Express the shaded area as the sum or difference of two integrals. DO NOT find the areas.



(d) Find the area enclosed between  $y = x^3 - 6x^2 + 8x$  and the  $x$ -axis.

3

**Question 4:**

- (a) Find the volume of the solid generated when the semicircle  $y = \sqrt{9 - x^2}$  is rotated about the  $x$ -axis.

3

- (b) (i) Show that the points A(0,3) and B(2,5) lie on the curve  $y = 3 + \sqrt{2x}$ .

2

- (ii) The part of the curve between A and B, is rotated about the  $y$ -axis. Find the volume of the solid generated.

4

**Question 5:**

- (a) Evaluate the definite integral  $\int x(\sqrt{x} + x^2)dx$ .

3

- (b) A rectangular pyramid is inscribed in a cylinder. The diameter of the base of the cylinder is 12cm and its height is 18cm. The sides of the rectangles are  $a$  cm and  $b$  cm.

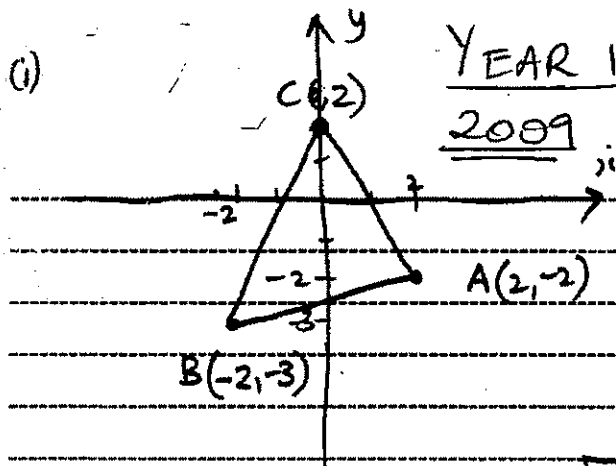
- (i) Show that the volume of the pyramid is  $6\sqrt{144a^2 - a^4}$

2

- (ii) Show that the maximum volume of the pyramid and the volume of the cylinder are in the ratio  $2:3\pi$ .

4

Q1 (i)



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TO

(ii) Length AC  $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$   
 $= \sqrt{(2 - 0)^2 + (-2 - 2)^2}$   
 $= \sqrt{4 + 16}$

$$d = \sqrt{20} \text{ units}$$

✓ (1)  $\sqrt{20} = 2\sqrt{5}$   
 $\therefore d = 2\sqrt{5}$

Gradient AC  $m = \frac{y_2 - y_1}{x_2 - x_1}$   
 $= \frac{-2 - 2}{2 - 0}$   
 $= \frac{-4}{2}$

$$\therefore m = -2$$

✓ (1)

(iii) Eqn AC  $y - y_1 = m(x - x_1)$

$$y - -2 = -2(x - 2) \quad (1) \quad \checkmark$$

$$y + 2 = -2x + 4$$

$$2x + y - 2 = 0$$

(2) ✓

(iv)  $D_p = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$

$$= \frac{|2 \cdot -2 + 1 \cdot -3 + -2|}{\sqrt{2^2 + 1^2}} \quad (1) \quad \checkmark$$

Point B(x, y) = (-2, -3)  
 $A = 2$   
 $B = 1$   
 $C = -2$

$$= \frac{|-4 - 3 - 2|}{\sqrt{5}}$$

$$= \frac{|-9|}{\sqrt{5}}$$

$$= \frac{9}{\sqrt{5}} \text{ u}^2 \quad (2) \quad \checkmark$$

(v)  $A = \frac{1}{2} \times b \times h$

$$= \frac{1}{2} \times \sqrt{20} \times \frac{9}{\sqrt{5}} \quad \checkmark$$

$$= \frac{1}{2} \times 2\sqrt{5} \times \frac{9}{\sqrt{5}}$$

$$A = 9 \text{ u}^2$$

(1)

7

$$Q1(b) \quad y = 3x^2 - 2x + 1$$

Eqn of tangent  $\frac{dy}{dx} = 6x - 2$

straight line  $y = 4x - 1$

here  $m = 4$  (1) ✓

$\therefore$  when  $y' = 4$ ,

that is the point where the gradients are parallel.

$\therefore 4 = 6x - 2$  (1) ✓

$x = 1$  sub into  $y = 3x^2 - 2x + 1$ .

$y = 3 - 2 + 1$   
 $= 2$

Point where tangent is parallel is (1, 2) (1) ✓

(3)

## QUESTION 2:

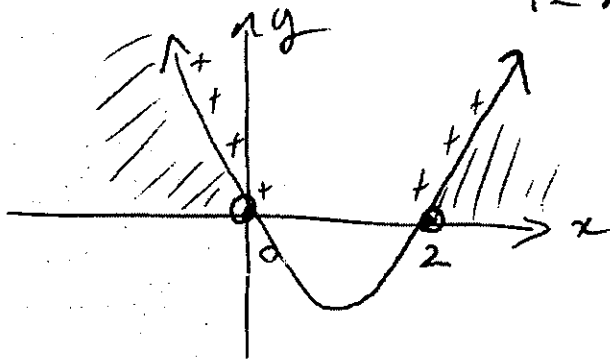
(a)  $f'(x) = 4x^3 - 12x^2$

$$f''(x) = 12x^2 - 24x \quad (1)$$

$\therefore$  For concave up,  $f''(x) > 0$

$$\therefore 12x^2 - 24x > 0$$

$$12x(x-2) > 0$$



$$\therefore \begin{aligned} x &> 2 \\ x &< 0 \end{aligned}$$

(1)

(b) (i)  $y = 3x^4 - 8x^3 + 6x^2$

$$\frac{dy}{dx} = 12x^3 - 24x^2 + 12x \quad (1)$$

Let  $\frac{dy}{dx} = 0$  for st. pts.

$$0 = 12x(x^2 - 2x + 1)$$

$$0 = 12x(x-1)(x-1)$$

$$\therefore x=0, x=1$$

when  $x=0, y=0$  (1)  
 $x=1, y=1$  (1)

(ii)  $\frac{d^2y}{dx^2} = 36x^2 - 48x + 12$  (1)  
 $\frac{d^2y}{dx^2} = 12(3x^2 - 4x + 1)$

Test  $x=0$ ,  $\frac{d^2y}{dx^2} = 12$  (1)  $\cup$  min.

Test  $x=1$ ,  $\frac{d^2y}{dx^2} = 0$  (1) horizontal  
 pt of inflex.

(iii) let  $\frac{d^2y}{dx^2} = 0$

$$0 = 12(3x - 1)(x - 1)$$

$\therefore x=1$  and  $x=\frac{1}{3}$ . (1)

For  $x=1$

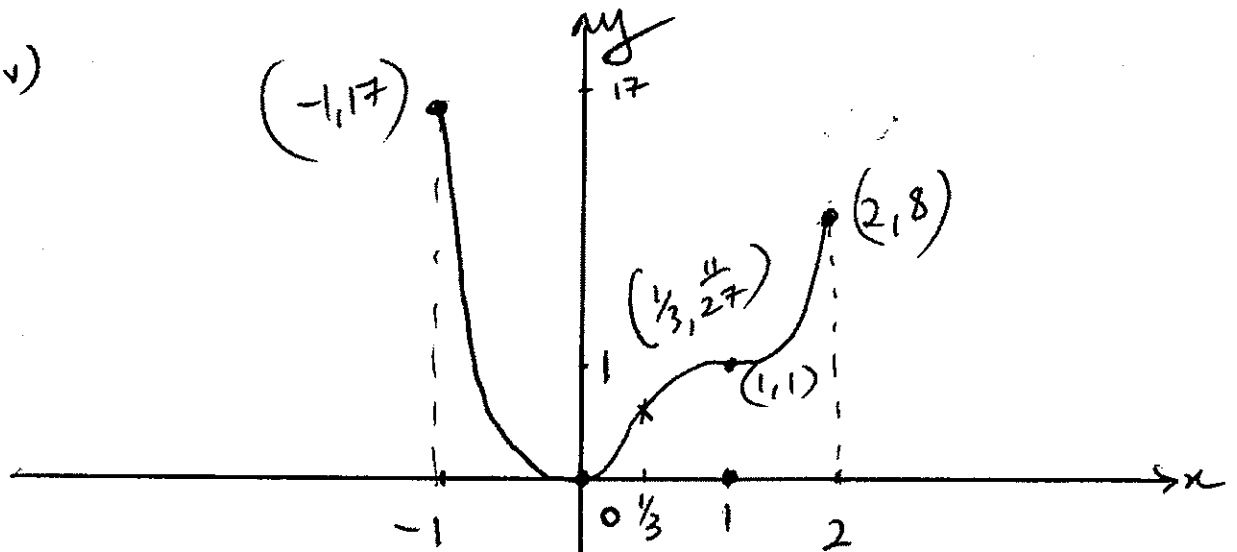
For  $x=\frac{1}{3}$

| $x$   | $\frac{1}{2}$ | 1 | 2 |
|-------|---------------|---|---|
| $y''$ | -             | 0 | + |

| $x$   | $\frac{1}{4}$ | $\frac{1}{3}$ | $\frac{1}{2}$ |
|-------|---------------|---------------|---------------|
| $y''$ | +             | 0             | -             |

(1)

(iv)



end pts ①  
showing st. pts ①  
general shape showing pt of inflexion ①

### QUESTION 3

(a) (i)  $\frac{3x^6}{6} - \frac{2x^3}{3} - \frac{x^2}{2} + C$  ①

$= \frac{x^6}{2} - \frac{2}{3}x^3 - \frac{x^2}{2} + C$  ①

(ii)  $\frac{3(3x-6)^5}{5 \times 3}$  ①

$= \frac{(3x-6)^5}{5} + C$  ①

$$(iii) \frac{d}{dx} = 2(x)^{-\frac{1}{2}} + 2x^{\frac{1}{2}} - 3 \quad (1)$$

$$= \frac{2x^{\frac{1}{2}}}{\frac{1}{2}} + \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} - 3x + C \quad (1)$$

$$= 4\sqrt{x} + \frac{4x\sqrt{x}}{3} - 3x + C \quad (1)$$

$$(b) \quad \frac{dy}{dx} = \frac{12(x-1)^3}{3} + C$$

$$\frac{dy}{dx} = 4(x-1)^3 + C$$

$$\text{when } \begin{cases} x=1 \\ \frac{dy}{dx}=3 \end{cases}$$

$$3 = 4(1-1)^3 + C \quad (1)$$

$$3 = C$$

$$\frac{dy}{dx} = 4(x-1)^3 + 3 \quad (1)$$

$$y = \frac{\cancel{4}(x-1)^4}{\cancel{4}} + 3x + C \quad \text{when } \begin{cases} x=1 \\ y=2 \end{cases}$$

$$2 = (1-1)^4 + 3 + C$$

$$-1 = C$$

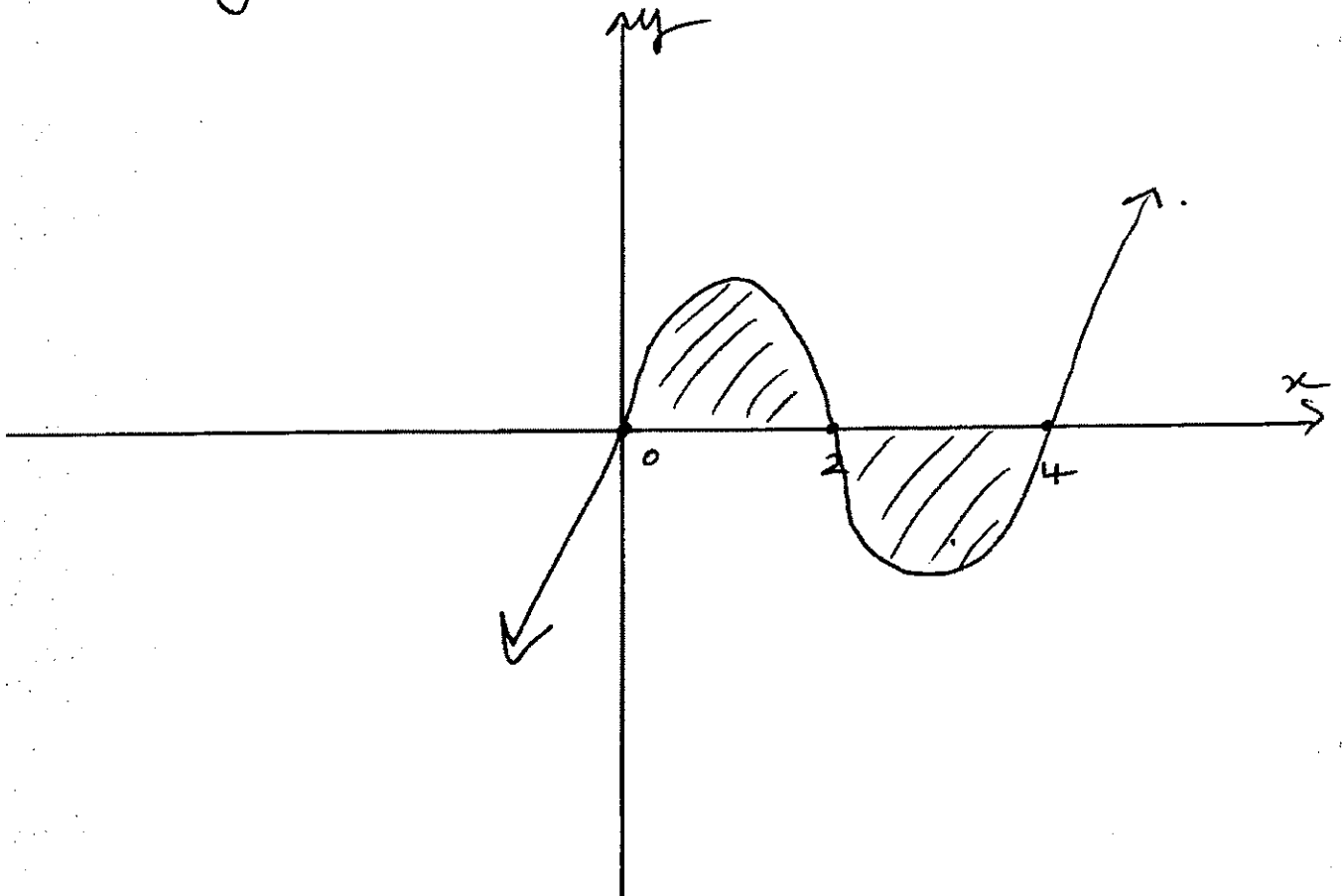
$$\therefore y = (x-1)^4 + 3x - 1 \quad (1)$$

$$(c) (i) A = \int_a^b \underset{\textcircled{1}}{f(x)} dx + \int_b^c \underset{\textcircled{1}}{g(x)} dx$$

$$(ii) A = \int_{-2}^2 \underset{\textcircled{1}}{\sqrt{4-x^2}} dx + \left| \int_2^4 \underset{\textcircled{1}}{-x+2} dx \right|$$

$$(d) \quad y = x(x^2 - 6x + 8)$$

$$y = x(x-2)(x-4)$$



$$\textcircled{1} \quad A = \int_0^2 x^3 - 6x^2 + 8x \, dx + \left| \int_2^4 x^3 - 6x^2 + 8x \, dx \right|$$

$$A = \left[ \frac{x^4}{4} - \frac{6x^3}{3} + \frac{8x^2}{2} \right]_0^2 + \left| \left[ \frac{x^4}{4} - \frac{6x^3}{3} + \frac{8x^2}{2} \right]_2^4 \right|$$

$$A = \left[ \left( \frac{16}{4} - 2 \times 8 + 16 \right) - (0) \right] + \left| \left[ \left( \frac{256}{4} - \frac{128}{1} + 64 \right) - \left( \frac{16}{4} - 16 + 16 \right) \right] \right|$$

$$A = 4 + \left| [(0) - (4)] \right|$$

$$A = 4 + 4$$

$$A = 8 \text{ units}^2 \quad \textcircled{1}$$

QUESTION 4:

$$(a) \quad V = \pi \int_{-3}^3 (\sqrt{9-x^2})^2 dx \quad (1)$$

$$V = \pi \int_{-3}^3 9-x^2 dx$$

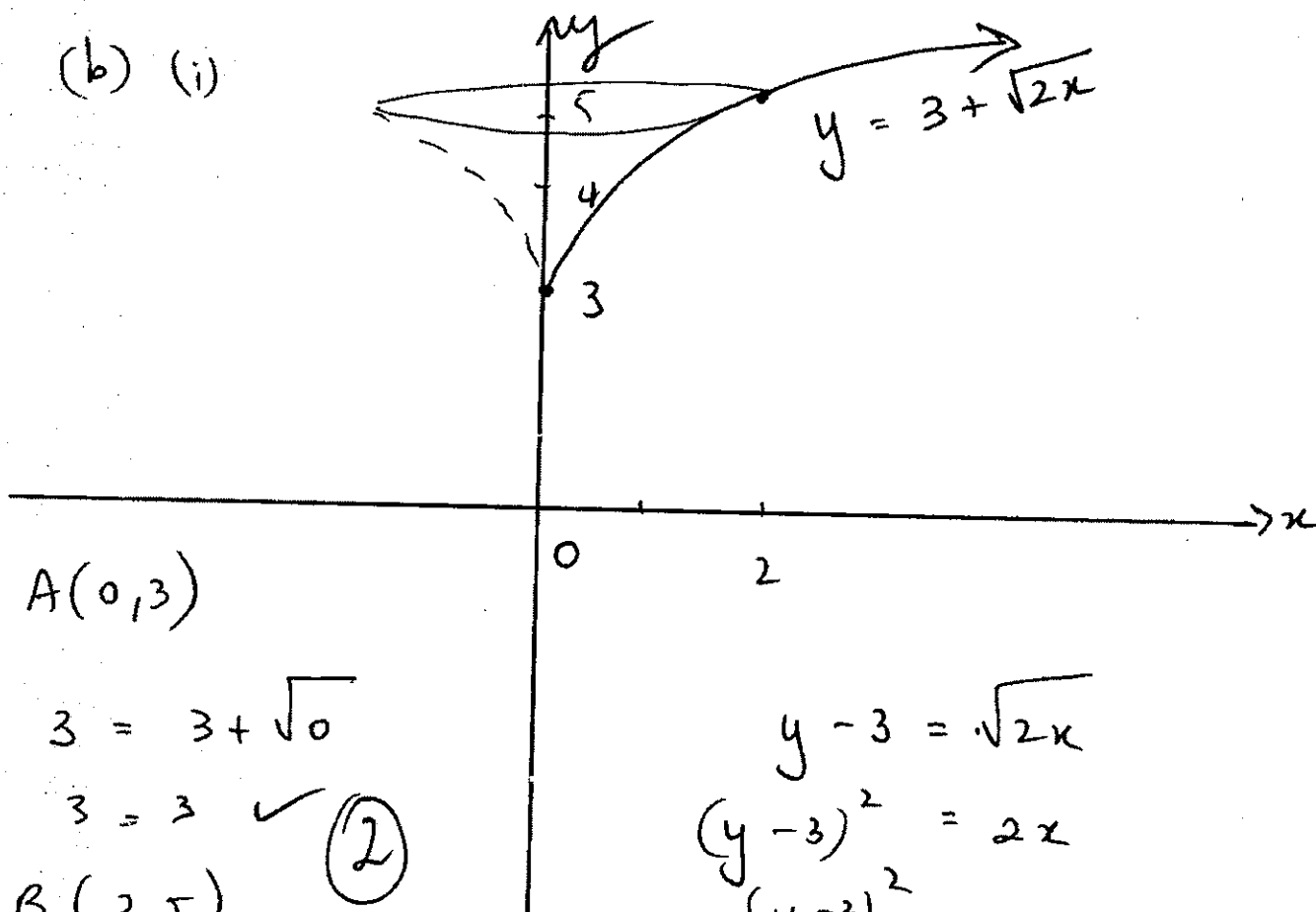
$$V = \pi \left[ 9x - \frac{x^3}{3} \right]_{-3}^3 \quad (1)$$

$$V = \pi \left[ \left( 27 - \frac{27}{3} \right) - \left( -27 + \frac{27}{3} \right) \right]$$

$$V = \pi [18 - (-18)]$$

$$V = 36\pi \text{ units}^3 \quad (1)$$

(b) (i)



$A(0, 3)$

$$3 = 3 + \sqrt{0}$$

$$3 = 3 \quad \checkmark \quad (2)$$

$B(2, 5)$

$$5 = 3 + \sqrt{2 \times 2}$$

$$5 = 3 + 2 \quad \checkmark$$

$$\begin{aligned} y - 3 &= \sqrt{2x} \\ (y - 3)^2 &= 2x \\ \frac{(y - 3)^2}{2} &= x \end{aligned}$$

(ii)

$$V = \pi \int_3^5 \frac{(y-3)^4}{4} dy \quad (1)$$

$$V = \pi \left[ \frac{(y-3)^5}{20} \right]_3^5 \quad (1)$$

$$V = \pi \left[ \frac{2^5}{20} - \frac{0}{20} \right] = \pi \left[ \frac{32}{20} \right] = \frac{8}{5} \pi u^3 \quad (1)$$

QUESTION 5:

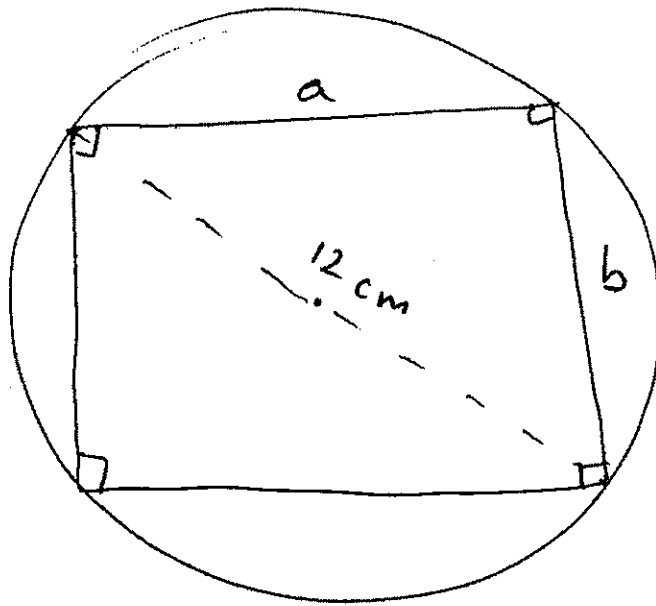
$$(a) \int_0^4 x^{3/2} + x^3 dx \quad (1)$$

$$= \left[ \frac{2x^{5/2}}{5} + \frac{x^4}{4} \right]_0^4 \quad (1)$$

$$= \left[ \left( \frac{64}{5} + 64 \right) - (0) \right]$$

$$= \frac{384}{5} \text{ OR } 76 \frac{4}{5} \quad (1)$$

(b) (i)



$$a^2 + b^2 = 144$$

$$b^2 = 144 - a^2$$

$$b = \sqrt{144 - a^2} \quad (1)$$

$$V \text{ of pyramid} = \frac{a \times b \times 18}{3}$$

$$V = \frac{a \times \sqrt{144 - a^2} \times 18}{3}$$

$$V = 6a \sqrt{144 - a^2}$$

$$V = 6 \sqrt{a^2(144 - a^2)} \quad (1)$$

$$V = 6 \sqrt{144a^2 - a^4}$$

$$(ii) \quad \frac{dv}{da} = \frac{1}{2} \times 6 (144a^2 - a^4)^{-\frac{1}{2}} \times (288a - 4a^3)$$

$$\text{let } \frac{dv}{da} = 0 \quad 0 = \frac{3(288a - 4a^3)}{(144a^2 - a^4)^{\frac{1}{2}}}$$

$$0 = 3(288a - 4a^3) \quad (1)$$

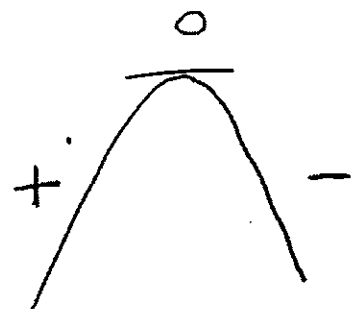
$$0 = 4a(72 - a^2)$$

$$0 = 4a(6\sqrt{2} - a)(6\sqrt{2} + a)$$

$$\therefore \boxed{a = 6\sqrt{2}} \quad (1)$$

| a               | 8 | $6\sqrt{2}$ | 9 |
|-----------------|---|-------------|---|
| $\frac{dv}{da}$ | + | 0           | - |

(1)



$$\begin{aligned} V \text{ of pyramid} &= \frac{6\sqrt{2} \times 6\sqrt{2} \times 18}{3} \\ &= 432 \text{ cm}^3 \end{aligned}$$

$$V \text{ of cylinder} = \pi \times 6^2 \times 18$$

$$= 648\pi \text{ cm}^3$$

$\therefore$

$$432 : 648\pi$$

$$2 : 3\pi$$

(1)